

CYCLES IN NATURE

Elements such as carbon, oxygen, hydrogen, nitrogen and sulphur are recycled in nature - this is the role of the decomposers.

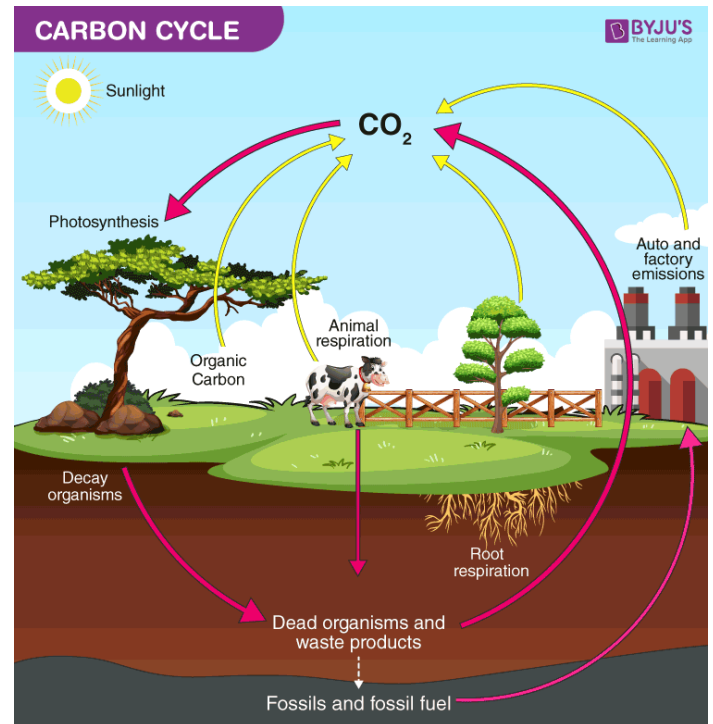
Carbon Cycle

The element carbon is one of the main materials in sugars, proteins and fats - these are found in the tissues of all living things. Carbon is also found in vitamins, as well as in waste products such as CO_2 and urea.

In the carbon cycle, plants convert CO_2 to food and O_2 during photosynthesis.

Animals (and plants at night) produce CO_2 during respiration.

On decomposition, all plant and animal material is reduced to CO_2 by the decomposers, and the residual nutrients are returned to the soil.



Nitrogen Cycle

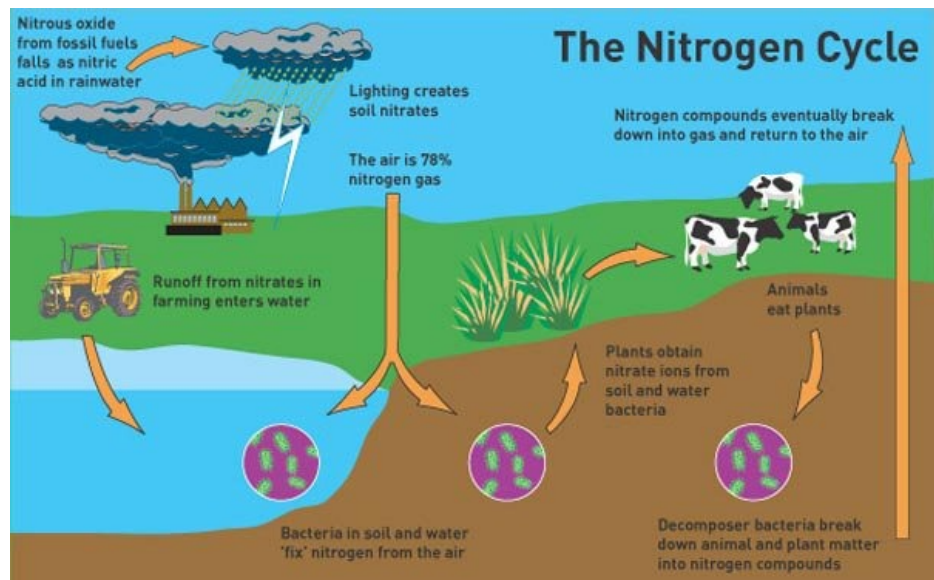
Nitrogen makes up 78% of the atmosphere, is found combined with oxygen as nitrates in the soil, and is one of the elements found in proteins (these are the building blocks of cells and tissues).

Animals obtain their nitrogen either from plants or other animals.

Plants get their nitrogen from the nitrates in the soil. They are soluble in water and are easily absorbed by plant roots.

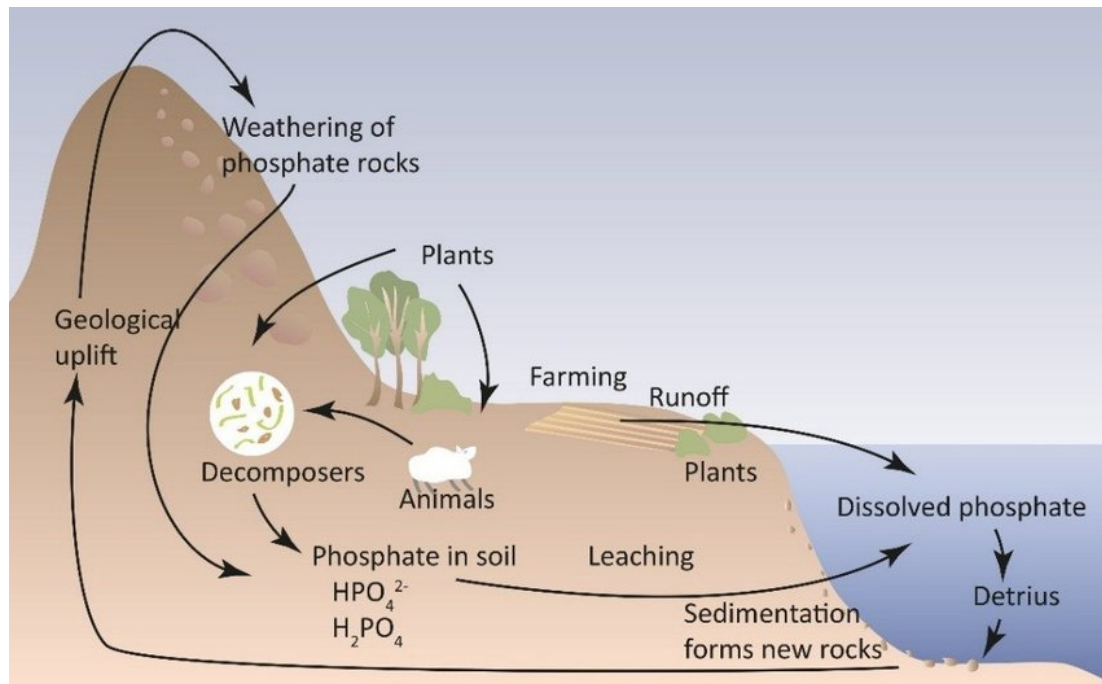
Nitrates can be formed in the following ways.

- The **decay of dead organisms** by decomposers.
- Atmospheric nitrogen changed by **nitrogen-fixing bacteria** in the soil and in **nodules** on the roots of **legumes** such as clover, peas and beans.
- **Lightning** causes atmospheric nitrogen N_2 to react with O_2 to form oxides of nitrogen. These form nitrates when they dissolve into water.



Phosphorus Cycle

The phosphorus cycle models how phosphorus moves from the lithosphere to the hydrosphere and then through food chains and back.



Phosphorus is essential for plant and animal growth, as well as the health of microbes inhabiting the soil, but is gradually depleted from the soil over time. The main biological function of phosphorus is that it is required for the formation of nucleotides, which comprise DNA and RNA molecules. Specifically, the DNA double helix is linked by a phosphate ester bond.

Calcium phosphate is also the primary component of mammalian bones and teeth, insect exoskeletons, phospholipid membranes of cells, and is used in a variety of other biological functions.

The phosphorus cycle is an extremely slow process, as various weather conditions (e.g. rain and erosion) help to wash the phosphorus found in rocks into the soil. In the soil, the organic matter (e.g. plants and fungi) absorb the phosphorus to be used for various biological processes.